DO NOW, AO1 REVISION

Conservation of energy, power and efficiency

Energy
Paper 1, Combined and Separate
22 questions

Cover the answers. Answer each question out loud or on paper, then check. Tick the ones you know. Learn the 7 starred (★) questions first.

	QUESTION	ANSWER	✓
1 ★	**Define power.**	The rate at which energy is transferred (or work is done).	☐
2 ★	**State the standard unit of power.**	Watt (W)	☐
3	**State the standard unit of time.**	Second (s)	☐
4	**State a unit that is equivalent to 1 W.**	1 J/s	☐
5	**State the term for energy transferred to stores that are not useful.**	Wasted energy	☐
6 ★	**State how unwanted energy transfers can be reduced.**	By lubrication and by thermal insulation.	☐
7	**Lubricating moving parts reduces unwanted energy transfers to which energy store?**	The thermal energy store	☐
8 ★	**Describe what it means for a device to be efficient.**	Most of the input energy is transferred usefully; little is wasted.	☐
9	**State the unit of efficiency.**	It has no unit (it is a ratio).	☐
10	**State how the efficiency of a mechanical energy transfer can be increased.**	By lubricating the moving parts.	☐
11	**State how the efficiency of a heating energy transfer can be increased.**	By adding thermal insulation.	☐
12	**Torches A and B each have an input power of 50 W. A has a useful output of 40 W and B of 45 W. Which is more efficient, and why?**	Torch B. Its efficiency is higher: 90%, against 80% for A.	☐

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	QUESTION	ANSWER	✓
13	**Heaters A and B each have a useful output of 2000 W. A has an input power of 2500 W and B of 2600 W. Which is more efficient, and why?**	Heater A. Its efficiency is higher: 80%, against 77% for B.	☐
14	**State the equation linking power, energy transferred and time.**	power = energy transferred ÷ time (P = E ÷ t)	☐
15	**State the equation linking power, work done and time.**	power = work done ÷ time (P = W ÷ t)	☐
16	**In words, state the equation for efficiency in terms of energy.**	efficiency = useful output energy transfer ÷ total input energy transfer	☐
17	**In words, state the equation for efficiency in terms of power.**	efficiency = useful power output ÷ total power input	☐
18 ★	**State what is meant by dissipated energy.**	Energy transferred to the surroundings and stored in less useful ways.	☐
19	**State what eventually happens to wasted energy.**	It is transferred to the surroundings, which warm up slightly.	☐
20 ★	**Two motors lift the same load through the same height. Motor A is quicker. Which is more powerful, and why?**	Motor A. It transfers the same energy in less time.	☐
21 ★	**Explain why efficiency can never be greater than 1 (100%).**	Energy cannot be created, so useful output cannot exceed total input.	☐
22	**A motor is 30% efficient. What does this mean?**	30% of the input energy is transferred usefully; 70% is wasted.	☐

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